

Learn something Sombaran

▼ IPC Shared Memory Algorithm Writer

- Initialize constants

```
name = "/my_shm"  
SIZE = 4096  
message = "Hello from writer process!\n"
```

- Open or create shared memory object

```
int shm_fd = shm_open(name, O_CREAT | O_RDWR, 0666);  
if (shm_fd == -1) {  
    perror("shm_open");  
    exit(EXIT_FAILURE);  
}
```

- Resize shared memory to desired length

```
if (ftruncate(shm_fd, SIZE) == -1) {  
    perror("ftruncate");  
    exit(EXIT_FAILURE);  
}
```

- Map shared memory into the process address space

```
void *ptr = mmap(NULL, SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, shm_fd, 0);  
if (ptr == MAP_FAILED) {  
    perror("mmap");  
    exit(EXIT_FAILURE);  
}
```

- Write data into the mapped region

```
memcpy(ptr, message, strlen(message) + 1);
```

- Cleanup local mappings and fd

```
munmap(ptr, SIZE);  
close(shm_fd);
```

▼ IPC Shared Memory Algorithm Reader

- Opens an existing named shared memory object `"/my_shm"` for read/write
- Maps it into the process address space
- Interprets the beginning of that memory as a C-string and prints it
- Cleans up (unmap + close FD)
- Optionally, it can remove the SHM object name via `shm_unlink(name)`